

- **Ratchet Floater:** A bond that pays a coupon that is a floating rate of interest, subject to coupon caps and floors that reset every period based on the current level of interest rates. As an example, such a bond may pay the following coupons:

Years 1–5: 3-month LIBOR + 15 bps, subject to the following:
 current coupon is at least as large as prior coupon;
 current coupon is no more than 25 bps above the
 prior coupon

- **Range Bond:** A bond that accrues a (high) coupon for each day some underlying index stays within a prespecified range and accrues a (low) coupon for each day the underlying index is outside of the prespecified range. These bonds are often callable. As an example, such a bond may pay the following coupons:

Years 1–5: 8%, accruing each day 3-month LIBOR
 is between 0% and 4%;
 0%, accruing each day 3-month LIBOR is above 4%

- **Leveraged Capped Floater:** A bond that behaves like a floating rate note when the underlying index is below a certain level and behaves like a leveraged inverse floater when the underlying index is above the predetermined level. As an example, such a bond may pay the following coupons:

Years 1–5: 6-month LIBOR + 50 bps, if 6-month LIBOR < 8%;
 $24.50\% - 2 \times \text{6-month LIBOR}$, if 6-month LIBOR
 $\geq 8\%$

The minimum coupon that this bond will pay is 0%, and the maximum coupon is 8.50%.

- **Snowball:** A bond that will appeal to investors who believe that interest rates will remain steady or decrease. The bond is structured such that the coupon goes up each period as a function of the prior period's coupon, provided that the interest rates remain steady or decline.⁷ Such bonds are often callable.

⁷The word “snowball” is based on a snowball rolling down a hill and increasing in size as it goes. Similarly, this bond is structured such that the coupons will increase in size as it goes, provided that rates remain the same or decline.

Year 1:	4%
Years 2–5:	Previous Coupon + 2.50% – 6-month LIBOR

The minimum coupon that this bond will pay is 0%.

- **Non-Inversion Notes:** This is also a bond whose coupons depend on the difference between two floating rate indexes. Often these bonds initially pay an (above-market) fixed rate coupon for some period of time. Thereafter, these bonds will pay either the above-market fixed coupon or zero, depending on whether or not the curve (as measured by the difference between the two floating rate indexes) remains above a predetermined level. Oftentimes, these bonds are also callable. As an example, such a bond may pay the following coupons:

Years 1–3:	6.00%
Years 3–10:	6.00%, accruing each day $\text{Curve} \geq 0$; 0.00%, otherwise
Curve:	10-year CMS – 2-year CMS
Call Feature:	This bond is callable in three years and every quarter thereafter

One thing to note is that a structured note (or any bond, for that matter) can never, ever pay a negative coupon. In many of the formulas above, there is specific mention of a minimum coupon that safeguards against a coupon formula that could take on negative values in certain instances. When we read the terms of a structured note, it is always implicitly understood that the coupons can never be negative even if this is not explicitly stated. As we will discuss in detail later on in this chapter, this sometimes necessitates that an option will need to be embedded into the note in order to guarantee that the coupon is never negative.

Another thing to note is that many of the bonds listed above are structured in a way such that they are callable by the issuer. Such a call feature is a valuable option and serves to increase the magnitude of the coupons that are paid in the note. Of course, the issuer will call the bond when it is advantageous for it to do so, which is usually at a time when it is to the disadvantage of the bondholder.

7.3 Size of the Market

Since the 1990s, the volume of newly issued structured notes has waxed and waned. Some types of structured notes have largely fallen out of favor with investors (e.g., leveraged inverse floaters), whereas others have remained popular with investors (e.g., range accrual notes). And in more recent years,

	2006	2007	2008	2009	2010
Range Accrual (includes Non-Inversion)	\$30,891	\$30,584	\$19,585	\$8,772	\$7,756
Fixed Rate Step-Up	5,918	3,667	7,835	42,559	56,267
Floating Step-Up	1,900	3,000			3,000
Capped Floater	2,914	2,240	700	2,750	5,400
Flipper	1,765	3,512	706	2,994	4,386
Callable Zero Coupon	5,005	8,759	7,038	12,985	21,030
Prime Linked	341				
Inverse Floater	3,379	3,603	1,633	625	986
Extendible	447	10,052	19,034	907	79
Other	25,650	34,195	22,061	67,022	73,767
Total	78,210	99,612	78,592	138,614	172,671

Table 7.1: USD Interest Rate-Linked Structured Note Issuance by Type (numbers in \$mm)

Source: Data courtesy mtn-i.

other types of notes such as non-inversion notes have become popular. Table 7.1 shows interest rate-linked structured note issuance by product type over the past number of years, as reported by mtn-i.

7.4 What Are Structured Notes?

Structured notes are bonds that are linked to some underlying derivatives transaction. These bonds can be issued by a variety of institutions, including banks, corporations, and certain government-sponsored enterprises (GSEs). The GSEs played a key role in the formation of the structured note market in the 1990s, but in more recent times other institutions, particularly banks, have emerged as the primary issuers of structured notes. In general, the issuers of structured notes have been relatively highly rated institutions, as investors who buy structured notes prefer to express the views that they are taking through the underlying derivatives transaction separately from their views on credit.

The key feature of a structured note is that it is created through an underlying swap transaction. In fact, the issuer of a structured note rarely retains any of the risks embedded in the bond. The issuer typically hedges out the risk inherent in paying out the coupons of the bond (whatever the nature

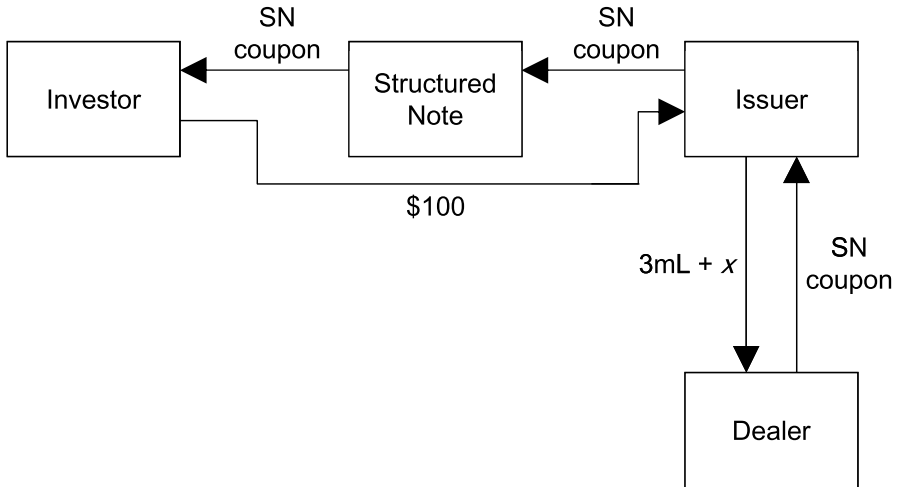


Figure 7.2: Issuer of a Structured Note Swapping the Coupons Back to LIBOR

of those coupons may be) by entering into an underlying swap transaction with a dealer.

This dynamic is shown in Figure 7.2. In this figure, an investor purchases a structured note issued by an issuer. The investor pays for the bond (assumed to be a par amount of \$100mm in this case) and will receive the structured note coupons over the life of the bond and principal at maturity. The issuer receives the purchase price for the bond when it is issued and is obliged to pay out the coupons in the structured note as well as principal at maturity. Bear in mind that the coupons in the structured note may be very complex; they may be leveraged inverse floating rate coupons, capped floating rate coupons, binary accrual fixed rate coupons, and so on.

As mentioned, the issuer usually does not maintain exposure to the structured note coupons and will enter into a swap transaction with a dealer in which the dealer pays the identical coupons to the issuer and the issuer usually pays a floating rate of interest, typically 3-month LIBOR plus or minus a spread. In this way, the issuer creates synthetic floating rate funding for itself and has completely hedged the risks inherent in paying the structured note coupons. Since the issuer is paying out the exact coupons that it is receiving from the dealer in the swap, it has no exposure to these cash flows. The issuer is left paying only the floating amount 3-month LIBOR + x bps over the life of the transaction. Often the synthetic floating rate funding obtainable by the issuer in a structured note transaction is significantly more

attractive (i.e., offers lower funding costs) than alternative funding opportunities. The dealer will earn bid/offer in the swaps transaction between itself and the issuer. This bid/offer compensates the dealer for the risks inherent in the swap transaction.

Dealers usually create separate structured note desks responsible for the marketing, structuring, underwriting, and the hedging and risk management of structured notes. This means that structured note desks need to enter into swaps and buy and sell options (often exotic in nature) from market makers of swaps and options, respectively. Although structured note desks may be captive customers of these other market making desks in their firm, others have found it advantageous to become customers of market making desks in swaps and options at other dealers. In our discussion, we will not draw such a distinction and just refer to the entity that enters into the underlying swap transaction in the creation of a structured note generically as a dealer.

How Do Structured Notes Come About?

In other words, why do the three entities (issuers, dealers, and investors) come together to create the transaction shown in Figure 7.2? There may be different potential drivers in this process, but most often it is a salesperson (or salespeople) at the dealer (or potentially at another firm altogether) who is the driver in the process.⁸ Salespeople may contact various investors to solicit interest in a particular structured note. If one or more investors is found who is interested in buying the proposed structured note, the dealer would arrange an acceptable issuer to issue the bond, typically off of some issuance vehicle such as a medium-term note shelf.⁹ Structured notes are relatively profitable transactions for the firms who arrange them, and salespeople involved in identifying buyers have been compensated accordingly.¹⁰

⁸Structured notes are sometimes sold by salespeople at so-called regional dealers who often do not have their own structured note desks and therefore work in conjunction with the desk at another dealer to create the note.

⁹It is important to note that an investor who purchases a structured note has exposure not only to the structured note coupons but also to the issuer as well. If the issuer were to go bankrupt and were unable to pay either coupons or principal in a structured note, the investor has no further recourse to the dealer who arranged the structured note transaction. Investors tend to prefer issuers of high creditworthiness when buying structured notes, thereby minimizing credit risk while often taking substantial interest rate risk in their purchases.

¹⁰At one point in time, it was common to have underwriting fees paid by issuers of structured notes to the firms who arranged those notes, and salespeople were presumably compensated based on these fees. In recent years, the payment of an upfront fee in a swap to recoup underwriting fees has been challenging for some issuers from an accounting perspective. Once the fee has come and gone at the inception of the swap, the remaining swap is no longer an $NPV = 0$ transaction and as such fails to qualify for certain preferred

It should be noted that there is a certain amount of “give and take” implicit in the situation depicted in Figure 7.2. The issuer will always want to pay the lowest funding costs possible, everything else the same. That is, for example, the issuer in Figure 7.2 would always prefer to pay funding costs of 3-month LIBOR – 25 bps instead of 3-month LIBOR – 10 bps. However, lower funding costs that are paid by the issuer to the dealer imply that the dealer will be willing to pay “lower” structured note coupons. That is, as the value of the cash flows that the dealer receives decreases, that will mean that the value of the cash flows that the dealer is willing to pay will decrease as well.¹¹ The “lower” structured note coupons will be passed on to the investor. There may come a point at which the investor is no longer interested in buying the bond if the value of the structured note coupons is decreased beyond a certain level. If this should be the case, either the issuer will have to be willing to live with higher funding costs or the dealer will have to be willing to make less bid/offer in the trade in order to create a note that will be attractive to the investor.

In summary, there are three degrees of freedom implicit in the transaction shown in Figure 7.2: the issuer’s funding costs, the investor’s coupons, and the dealer’s bid/offer. Any of these may be altered so as to provide greater or lesser value to any of the three constituents in the trade. Table 7.2 summarizes the net cash flows to and from each of these participants over the life of the trade.

Although the issuer is economically unaffected by the structured note coupons that are paid to the investor, this does not imply that the issuer is necessarily completely passive regarding their nature. The complex nature of structured notes means that investors who buy them, particularly retail investors, sometimes have limited ability to value these notes. Due to reputational effects associated with its name, an issuer typically wants to make sure that the bond it is issuing is being sold at a price that is reasonably in line with its value. That is, if it is subsequently determined by investors that the difference between the price the bond was sold for and its true value was egregious, this could significantly hamper the ability of the issuer to sell further such notes to investors. This dynamic helps keep the magnitude of the bid/offer in the underlying swap transaction and the

accounting treatment under FAS 133. It has therefore become more commonplace for salespeople to be compensated by receiving credit for a portion of the bid/offer earned in the underlying swap transaction, rather than through an underwriting fee.

¹¹The dealer is earning a certain amount of bid/offer in the swap transaction. If the funding costs of the issuer decrease, the dealer could possibly maintain the same structured note coupons if it were willing to forgo some of its bid/offer in the trade.

	Investor	Issuer	Dealer
Net Cash Flow At Inception	– Principal	+ Principal	
Net Cash Flows Over Life	+ SN Coupons	– LIBOR Coupons	+ LIBOR Coupons – SN Coupons
Net Cash Flow At Maturity	+ Principal	– Principal	
Benefits	Customized Note	Attractive Funding	Bid/Offer in Swap

Table 7.2: Cash Flows and Benefits of Structured Note Transaction by Participant

synthetic floating rate funding obtainable by the issuer within reasonable limits.

How could an investor become aware that egregious fees were taken out of the bond at inception? After the bond is sold, the dealer who arranged for the offering will often have to send periodic mark-to-market levels on the bond, providing an indication of the bond's value in the current market environment. If the mark on the bond is too far below its issue price, particularly soon after it has been issued, this could call into question the initial pricing of the bond. Moreover, on occasion investors will want to sell structured notes out of their portfolios. The secondary market for structured notes is not very liquid to begin with, and excessive fees taken out of these trades at their inception could impact the bid that an investor sees if he tries to sell the note in the secondary market.

One final point worth emphasizing is that investors who buy structured notes are invariably taking a view against the realization of the forward curve. A structured note buyer is buying the bond because he believes it to be an investment vehicle through which he can express his view on the market, whether that view has to do with the direction of rates, the shape of the curve, or the level of implied volatility. Structured notes are created in accordance with the forward curve; that is, the coupons paid by the issuer are valued based on the forward curve obtained through bootstrapping since these are rates that can be hedged. Inasmuch as the forwards are wrong, and the actual realization of rates is different from that implied by the forward curve, the buyer of a structured note may end up with an enhanced return.

7.5 In the Beginning . . . Floating Rate Notes

In the early days of the structured note market, the most popular bonds created were floating rate notes. These are bonds whose coupons are not fixed but reset periodically off of some floating rate index such as LIBOR, Prime, T-Bills, Fed Funds, and CMT. These bonds are not terribly exciting in terms of their derivatives complexity. But, nonetheless, this category of structured notes is where it all began.

Floating rate notes became very popular among investors who were concerned that rates would rise. In a rising rate environment, fixed rate bonds lose their value. But floating rate notes tend to retain their value in a rising interest rate environment. They have very little interest rate exposure, and the reset mechanism tends to bring their value back near par on any reset date.

7.5.1 A Prime Floating Rate Note

A *Prime floating rate note* is a floating rate note whose coupons are based on Prime. As an example, let's consider the Prime floating rate note whose terms are shown in Table 7.3.¹² Prime is a floating rate of interest that banks charge to their middle-market clients. Figure 7.3 shows the relationship between Prime and LIBOR in the late 1980s and early 1990s. There are two features about Prime that are worth noting. First, some people in the market have suggested that Prime is a “sticky” index. Referring to Figure 7.3, what this means is that during periods of time in which LIBOR is falling, Prime will fall as well, but the rate of descent of Prime may be more gradual. From the point of view of a floating rate note buyer, any stickiness in Prime would give it a relative advantage over LIBOR-based notes in a falling rate environment. Whether or not the stickiness of Prime is an accurate statement, it was a selling point that structurers would use when trying to discuss the relative advantages of a Prime floater over a LIBOR floater.

The second feature worth noting is that the standard convention for Prime floating rate notes is that the Prime rate has a daily (weighted) reset mechanism. The daily reset mechanism indicates that we will look at each of the Prime settings daily throughout a period, average those settings, and pay the average at the end of the period.¹³ This daily reset mechanism is a

¹²For a more detailed analysis of Prime floating rate notes, see Peng and Dattatreya (1995).

¹³The daily weighted reset mechanism found in Prime floating rate notes concurs with the standard convention on the Prime leg of a Prime-LIBOR basis swap, which also has a daily weighted reset mechanism. Whereas the standard convention for LIBOR in a swap is that we set the rate once at the beginning of the period, accrue that single rate, and

Issuer:	U.S. Agency
Principal:	\$100mm
Maturity:	2 years
Coupon:	Prime – 2.80%
Day Count:	Act/360
Reset and Payment Dates:	Prime resets daily and is paid quarterly
Issue Price:	100%

Table 7.3: Terms of a Prime Floating Rate Note

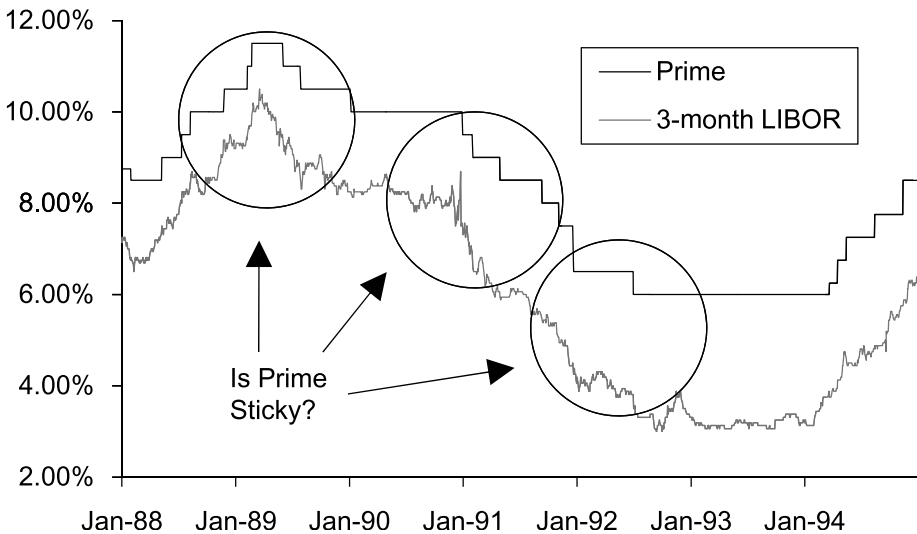


Figure 7.3: The Relationship Between Prime and LIBOR

Source: Figure adapted from Peng and Dattatreya (1995).

good feature in a rising rate environment. If rates are rising, the daily reset mechanism allows the holder of a Prime note to participate in the higher rate settings over the period. In a LIBOR floater, since we set the rate once at the beginning of each period, if rates were to rise in a period after the

pay it at the end of the period, the Prime leg uses an average of rates over the period. The word *weighted* refers to the fact that the reset for Friday counts for both Saturday and Sunday as well.

initial rate was set, the holder of the note would not get to participate in the higher rates for the remainder of that period.

How Did We Get the Coupon of Prime – 2.80%?

Let's refer back to Figure 7.2 and keep the dynamics of structured note issuance in mind in order to figure out how the coupon in this, and any, structured note was determined. The coupon in the note was determined by the economics implicit in the underlying swap transaction between the dealer and the issuer. In this case, the dealer is going to be paying Prime-based coupons to the issuer, and the issuer will be paying LIBOR-based coupons to the dealer in order to create synthetic LIBOR-based funding for itself.

Thus, it is a Prime-LIBOR basis swap that the dealer and the issuer have transacted in order to create this note. The economics of this basis swap will determine the ultimate coupon that can be paid out in the note. Suppose that the issuer indicates that it is willing to fund itself at 3-month LIBOR – 25 bps for two years. Suppose further that mid-market in a 2-year Prime-LIBOR basis swap is Prime – 250 bps versus 3-month LIBOR flat. Suppose further that the dealer is willing to trade the basis swap 5 bps from mid-market on either side. Thus, the dealer is willing to pay Prime – 255 bps versus receiving 3-month LIBOR for two years. As a reminder, the standard convention in the Prime-LIBOR basis swap is that Prime is reset daily (weighted) and paid quarterly Act/360. Since the issuer will actually be paying 3-month LIBOR – 25 bps instead of 3-month LIBOR flat, this means that the dealer would be willing to pay Prime – 255 – 25, or Prime – 280 bps.¹⁴

¹⁴Think about it: if the dealer is willing to pay Prime – 255 versus 3-month LIBOR, then by “subtracting” 25 bps from each side that must mean the dealer is willing to pay Prime – 280 versus 3-month LIBOR – 25.

By the way, note that with a coupon of the form Prime – 280 bps, it is theoretically possible for this quantity to be negative. This would occur if Prime were to be lower than 2.80%. As mentioned previously, when we see a coupon like Prime – 280 bps, it is implicitly understood that there is a coupon restriction implicit, and the minimum coupon is zero. As will become clearer later in this chapter, in order to maintain a minimum coupon of zero, it is (technically) necessary to embed a floor on Prime struck at 2.80%. However, when the structured note market first took off, there was not a vibrant market for such floors. With an eye to Figure 7.3, most dealers viewed the chances of Prime going below 2.80% as essentially zero. Dealers often wrote these options for free (or a negligible premium), willing to live with the consequences if they were wrong (meaning, they would have to provide the payments on this in-the-money floor, without having collected an upfront premium for its sale). During the 1990s when these notes were popular, Prime never set low enough for this issue to be of consequence.